PUMF Insights Assistant

Reference architecture on AWS and Google Cloud, with estimated monthly costs in CAD

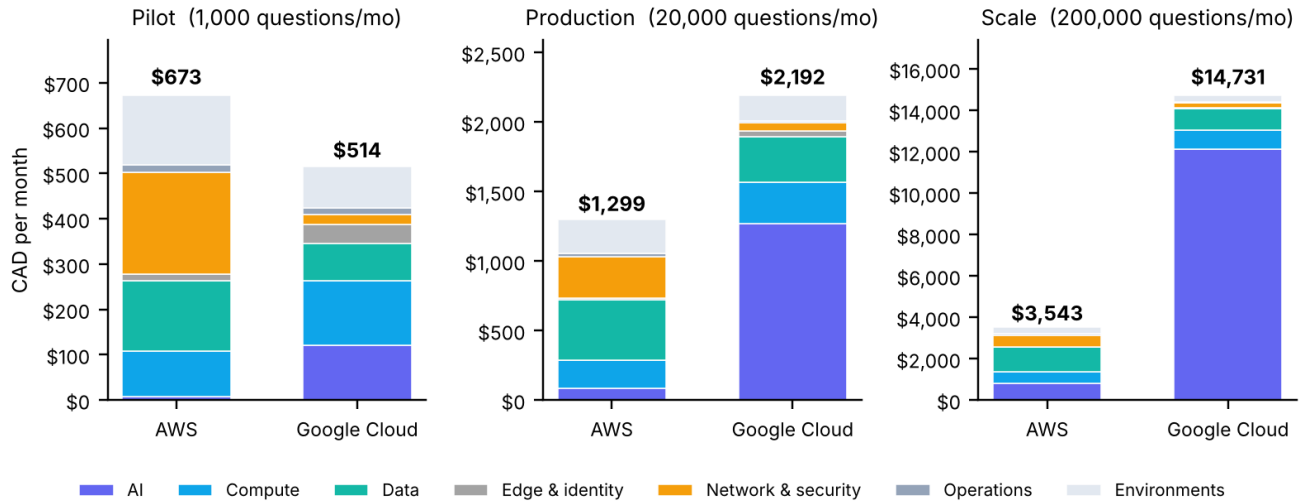
Prepared November 9, 2026. Canadian regions: AWS ca-central-1 and Google Cloud northamerica-northeast2 (Toronto). Prices are list prices converted at USD 1 = CAD 1.41.

Summary

Both clouds run the same design: an LLM translates a planner's question into a validated query spec, and a deterministic DuckDB service computes weighted estimates from StatCan PUMFs. At production volume (20,000 questions a month) the estimate is **$1,299/month on AWS** and **$2,192/month on Google Cloud**, all-in including dev and test.

The gap is driven by the model, not the infrastructure. Keeping inference in Canada limits each cloud to one model family today: Claude Haiku 5.5 through Bedrock's Canada cross-Region inference, or Gemini 3.5 Flash on Vertex AI's Montreal endpoint. Haiku 5.5 costs roughly a tenth as much per token. At pilot volume, fixed infrastructure dominates and Google Cloud is cheaper; from production volume upward, model tokens dominate and AWS is cheaper.

Tier	Questions/mo	Users	AWS CAD/mo	Google Cloud CAD/mo	AWS per question	GCP per question
Pilot	1,000	25	$673	$514	$0.672	$0.514
Production	20,000	200	$1,299	$2,192	$0.065	$0.110
Scale	200,000	2,000	$3,543	$14,731	$0.018	$0.074



Monthly cost by layer. Per-question figures divide the total by user questions only; 1,000 evaluation-set runs a month are included in every tier's costs.

Workload assumptions

Assumption	Pilot	Production	Scale
User questions per month	1,000	20,000	200,000
Active users	25	200	2,000
Evaluation-set runs per month	1,000	1,000	1,000

Assumption	Pilot	Production	Scale
Result-cache hit rate (skips tabulation)	15%	30%	40%
Tokens per question (4 LLM calls, +10% clarify)	~16k in / ~2k out	same	same
Warm tabulation instances	1	2	5
Catalog database	1 ACU / 1 vCPU, no HA	2 ACU + reader / 2 vCPU HA	4 ACU + 2 readers / 4 vCPU HA + replica
Dev + test environments	30% of prod infra	25%	15%

Service mapping

Each logical component has a direct equivalent on both clouds, so the application code (Python, DuckDB, PostgreSQL with pgvector, JSON query specs) is portable. Only orchestration definitions, IAM and the model client differ.

Component	AWS (ca-central-1)	Google Cloud (Toronto)
Edge, WAF	CloudFront, AWS WAF	External HTTPS load balancer, Cloud Armor
Identity	Cognito (federate to Entra ID / Okta)	Identity Platform or IAP
API	API Gateway HTTP API	Load balancer to Cloud Run
Orchestration (runtime)	Step Functions Express	Workflows
Step logic	Lambda (Python)	Cloud Run services (Python)
Tabulation engine	Lambda container, DuckDB, provisioned concurrency	Cloud Run, DuckDB, min instances
LLM	Bedrock: Claude Haiku 5.5, Canada inference profile	Vertex AI: Gemini 3.5 Flash, Montreal endpoint
Embeddings	Titan Text Embeddings V2	Vertex AI text embeddings
Variable catalog	Aurora PostgreSQL Serverless v2 + pgvector	Cloud SQL PostgreSQL + pgvector
Data lake zones	S3 + SSE-KMS, Object Lock on landing	Cloud Storage + CMEK, retention lock
Physical catalog	Glue Data Catalog, Lake Formation	Dataplex Universal Catalog, BigLake
Ingestion	EventBridge, Step Functions Standard, Glue	Eventarc, Workflows, Cloud Run jobs
Data quality	Glue Data Quality	Dataplex data quality
Result cache	DynamoDB on-demand	Firestore
Analyst SQL, answer log	Athena over S3	BigQuery
Private networking	VPC interface endpoints (no NAT)	Direct VPC egress, Private Google Access, VPC-SC
Keys and secrets	KMS, Secrets Manager	Cloud KMS, Secret Manager
Audit, threat detection	CloudTrail, GuardDuty, Security Hub, Config	Cloud Audit Logs, Security Command Center
Observability	CloudWatch, X-Ray	Cloud Logging, Monitoring, Trace

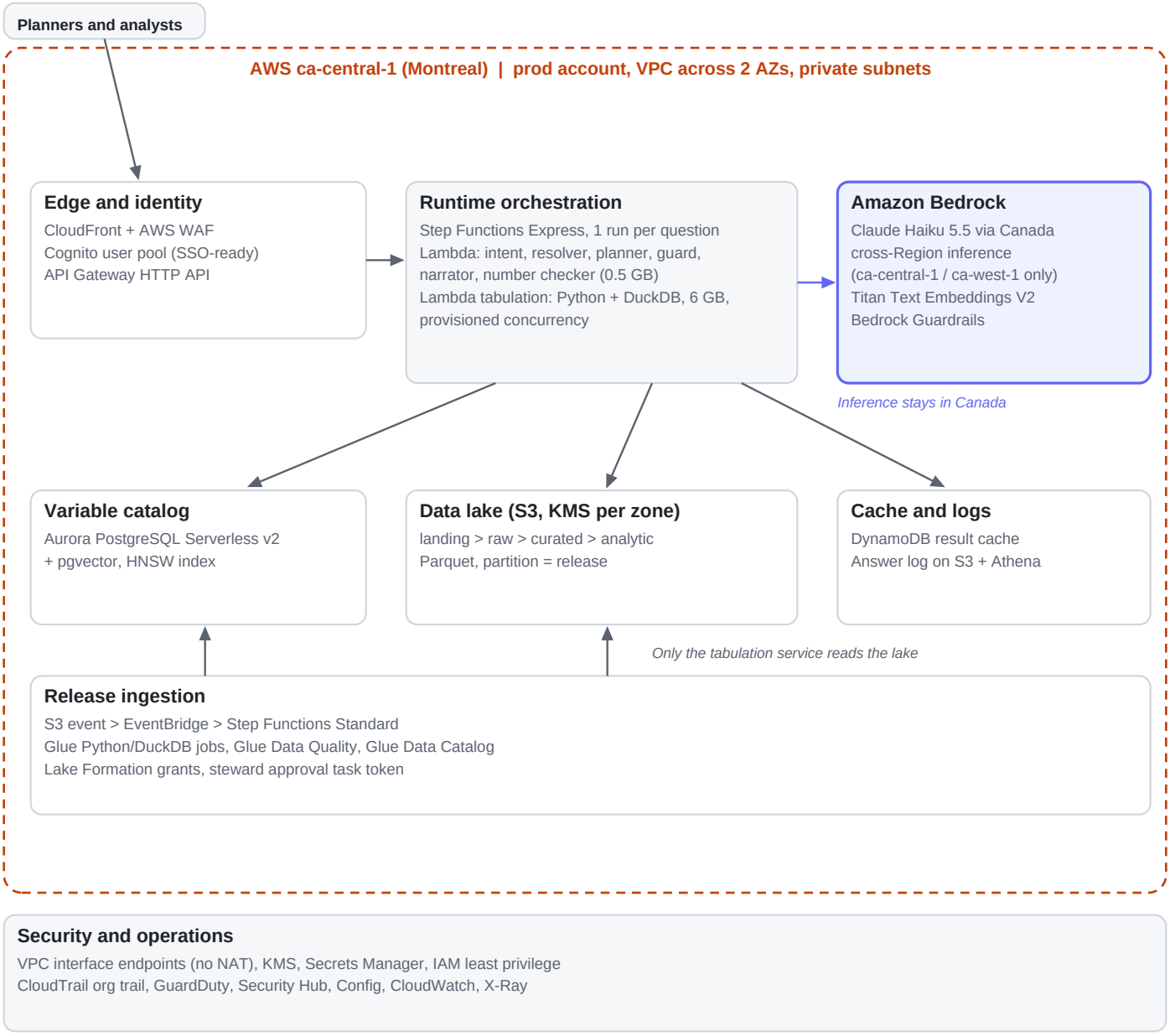
Design constants on both clouds

- The LLM never produces a number: it emits a JSON query spec validated against the catalog; DuckDB computes every estimate.

- No identity that calls the LLM can read microdata; the model sees catalog metadata and finished, reliability-checked tables only.

- Releases are immutable and versioned; reliability rules from each user guide run before the narrator; all data and inference stay in Canada.

AWS architecture



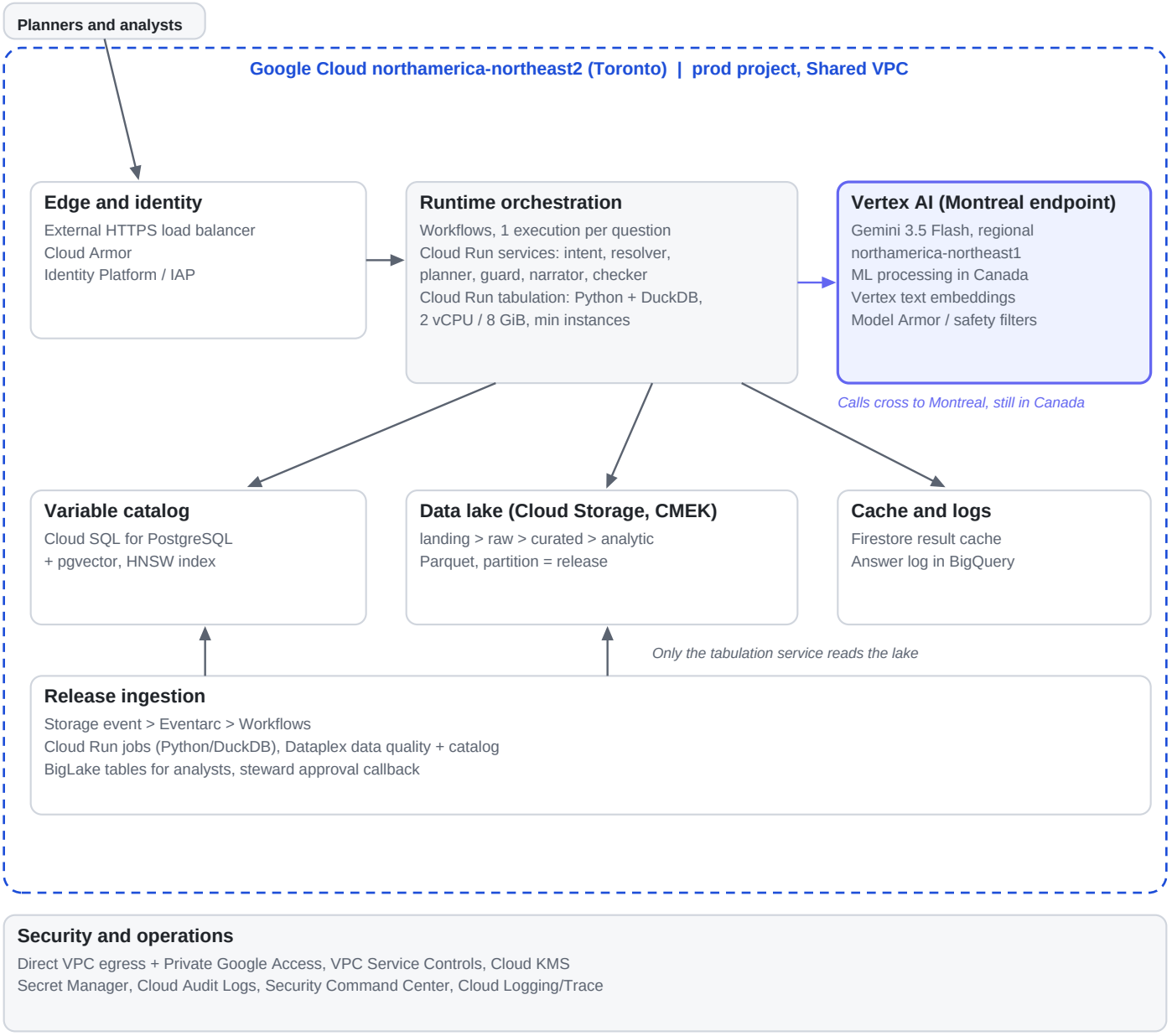
- Bedrock's Canada cross-Region inference for Claude Haiku 5.5 routes only between ca-central-1 and ca-west-1, so prompts never leave Canada. Larger Claude models are reachable from ca-central-1 only through US or global routing.

- Seven interface endpoints across two Availability Zones cost about CAD 160/month regardless of volume; they replace a NAT gateway and keep traffic private.

- Aurora Serverless v2 is the largest fixed cost. Dev and test clusters should use auto-pause (0 ACU).

- Provisioned concurrency keeps the 6 GB tabulation function warm so DuckDB answers in under 3 seconds.

Google Cloud architecture



- Gemini 3.x models are offered on global, US and EU endpoints; Gemini 3.5 Flash is listed with ML processing in northamerica-northeast1 (Montreal). The Toronto workloads call the Montreal endpoint, so inference still stays in Canada.

- Toronto is a Tier 2 region for Cloud Run, about 40% above Tier 1 prices; compute is still a small share.

- Direct VPC egress and Private Google Access remove the per-endpoint hourly fees AWS charges.

- Cloud SQL has no scale-to-zero; the pilot uses a 1 vCPU instance without HA, adding HA at production.

Cost detail

CAD per month, production account plus dev and test. Basis: S = unit price confirmed from a November 2026 source; L = published US list price, scaled by a stated regional factor; A = planning allowance.

AWS

Layer	Service	Basis	Pilot	Production	Scale
AI	Bedrock: Claude Haiku 5.5, Canada cross-Region inference	S	$8	$86	$823
AI	Bedrock: Titan Text Embeddings V2 (queries + re-embeds)	L	$0	$0	$2
Compute	Lambda: orchestration + tabulation	L	$1	$8	$68
Compute	Lambda provisioned concurrency (1 x 6 GB)	L	$93	$185	$463
Compute	Step Functions Express (runtime) + Standard (ingest)	L	$0	$1	$6
Compute	Glue jobs + Data Quality (ingest, amortized)	L	$5	$5	$5
Data	Aurora PostgreSQL Serverless v2 + pgvector	S	$153	$425	$1,104
Data	S3: landing, raw, curated, analytic, exports	L	$1	$2	$9
Data	DynamoDB on-demand: result cache	L	$0	$0	$1
Data	Athena: analyst SQL + answer-log queries	L	$2	$8	$78
Edge & identity	CloudFront + API Gateway HTTP API	L	$0	$0	$2
Edge & identity	AWS WAF (1 web ACL, 5 rules)	L	$14	$14	$15
Edge & identity	Cognito user pool	L	$0	$0	$0
Network & security	VPC interface endpoints (7 x 2 AZ)	L	$159	$159	$159
Network & security	KMS (5 keys) + Secrets Manager (6)	L	$11	$12	$27
Network & security	GuardDuty, Security Hub, Config, CloudTrail	A	$56	$127	$352
Operations	CloudWatch logs, metrics, alarms; X-Ray	L	$17	$23	$75
Environments	Dev + test accounts (30% of non-AI prod)	A	$153	$242	$355
Total			**$673**	**$1,299**	**$3,543**

Google Cloud

Layer	Service	Basis	Pilot	Production	Scale
AI	Vertex AI: Gemini 3.5 Flash, Montreal regional endpoint	S	$121	$1,265	$12,112
AI	Vertex AI: text embeddings (queries + re-embeds)	A	$1	$2	$13
Compute	Cloud Run services: orchestration + tabulation	L	$2	$19	$174
Compute	Cloud Run min instances (1 x 2 vCPU/8 GiB, idle)	L	$130	$259	$648
Compute	Workflows (runtime + ingest)	L	$1	$10	$99
Compute	Cloud Run jobs + Dataplex data quality (ingest)	L	$9	$9	$9
Data	Cloud SQL PostgreSQL + pgvector (1 vCPU/3.75 GiB)	S	$82	$324	$952
Data	Cloud Storage: landing, raw, curated, analytic, exports	L	$1	$2	$9
Data	Firestore: result cache	L	$0	$0	$1
Data	BigQuery: analyst SQL + answer log	L	$0	$0	$80
Edge & identity	External HTTPS load balancer	L	$28	$28	$29
Edge & identity	Cloud Armor (1 policy, 5 rules)	L	$14	$14	$15
Edge & identity	Identity Platform / IAP	L	$0	$0	$0
Network & security	Direct VPC egress + Private Google Access	L	$0	$0	$0
Network & security	Cloud KMS (5 keys) + Secret Manager (6)	L	$1	$3	$18
Network & security	Security Command Center, audit logs, Org Policy	A	$21	$56	$212
Operations	Cloud Logging, Monitoring, Trace	L	$14	$14	$21
Environments	Dev + test projects (30% of non-AI prod)	A	$91	$185	$340
Total			**$514**	**$2,192**	**$14,731**

Model choice is the biggest lever

LLM cost alone, CAD per month, for the same token volumes. Options marked 'leaves Canada' send prompts to US or global endpoints.

Model option	Residency	Pilot	Production	Scale
Claude Haiku 5.5, all four calls (costed)	Canada	$8	$86	$823
Haiku 5.5 + Claude Sonnet 5.5 for planner and narrator	Leaves Canada	$84	$886	$8,483
Gemini 3.5 Flash, Montreal (costed)	Canada	$121	$1,265	$12,112
Gemini 3.8 Flash, global, 2027 standard rate	Leaves Canada	$112	$1,173	$11,223

Risks, levers and open questions

Model quality on Haiku 5.5. The AWS estimate assumes Haiku 5.5 handles query planning well. Run the golden question set before committing; if planning accuracy falls short, routing only the planner to a larger model changes residency and roughly ten-folds LLM cost.

Regional and promotional pricing. Canada inference-profile pricing is assumed at the 10% geographic premium reported for Bedrock regional endpoints. Gemini prices on regional endpoints and any 2027 changes should be confirmed with Google.

Unverified list prices. Lines marked L use published US list prices with a 10% regional factor (40% for Cloud Run Tier 2). Validate them in the AWS Pricing Calculator and Google Cloud Pricing Calculator before budgeting.

Commitments. Database Savings Plans (AWS) and committed use discounts for Cloud SQL and Cloud Run (Google) typically cut the fixed infrastructure lines by 20 to 50% at production volume.

Prompt caching. The intent, rerank and planner prompts share long, stable instructions and catalog context. Prompt or context caching on either cloud can cut input-token cost substantially; not included here.

Excluded. Staff time, support plans, egress beyond free tiers, taxes, domain and certificate costs, and statistical matching work for a fused CIUS-Census file.

Sources

Amazon Bedrock now offers Canada cross-Region inference for Claude Haiku 5.5 (Oct 30, 2026)

AWS Bedrock pricing 2026, CloudForecast (Haiku 5.5, Sonnet 5.5 rates)

AWS Bedrock pricing 2026, Economize (regional endpoints +10%)

Claude in Amazon Bedrock: endpoint types by region

Aurora Serverless v2 pricing guide 2026, usage.ai

Aurora Serverless v2 supported regions (Canada Central)

Gemini API pricing, May 2026 (Gemini 3.5 Flash rate)

Gemini API pricing 2026 (Gemini 3.8 Flash and 2027 rates)

Google developer forum: Canada ML processing for Gemini 3.5 Flash

Cloud Run locations and pricing tiers

Cloud SQL pricing

Cloud SQL C4A regional prices, Spare Cores (Toronto vs Iowa)

Bank of Canada daily exchange rates